Olivia Curtis CURRICULUM VITÆ

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ACADEMIC POSITIONS

November 2024 – Present
The Pennsylvania State University, *University Park, PA*
PSETI Center Postdoctoral Fellow
Department of Astronomy and Astrophysics
The Penn State Extraterrestrial Intelligence Center
Research Advisor: Jason T. Wright

EDUCATION

May 2020 – August 2024
Boston University, *Boston, MA*
Doctor of Philosophy, Astronomy
Dissertation Title: Properties of Voids and Void Galaxies
Research Advisor: Tereasa G. Brainerd

September 2018 – May 2020
Boston University, *Boston, MA*
Master of Arts, Astronomy
Graduate Coursework: Galaxies, High-energy Astrophysics, Plasma Physics, Radiative Transfer, Fluid Mechanics

August 2014 – May 2018
University of South Florida, *Tampa, FL*
Bachelor of Science, Physics
Bachelor of Arts, Mathematics (pure concentration)
USF Honors College
GPA: 3.98/4.00, Summa Cum Laude
Graduate Coursework: Computational Physics, General Relativity

GRANTS

[1] 2026 CO-I – Heising-Simons Foundation Science Events and Gatherings "Summer Astronomy Training for Undergraduate Research and Networking (SATURN) Workshop" – $60,000
[2] 2026 CO-I – Nancy Grace Roman Characterizing the Interstellar Medium with Stacked Spectra of Ten Million Galaxies – $150k
[3] 2026 CO-I – Nancy Grace Roman A Census of Black Hole Evolution using High-Ionization Emission Lines – $150k
[4] 2026 CO-I – JWST Cycle 5 "CEDAR: Characterizing Electron Densities At Reionization" – $234,113
[5] 2025 CO-PI – PSU Institute for Computational and Data Science Rising Researcher Grant – $75k
[6] 2023 Graduate School of Arts & Sciences Summer Fellowship – $6,375.50
[7] 2023 BU Graduate Student Organization Travel Grant – $1,000
[8] 2017 Society of Physics Students National Travel Grant – $1,000
[9] 2017 USF Travel Grant – $1,000

Proposals Under Review

[1] 2026 PI – CHiwi Foundation Space Exploration Science Grants Program "When Galaxies Glow with Industry: The Search for Galaxy-Spanning Civilizations Harnessing Energy from Black Holes" – $123,482

[2] 2026 CO-PI – Alfred P. Sloan Foundation Matter-to-Life Seed Grant "When Galaxies Glow with Industry: Disentangling Dyson Sphere Waste Heat from Inorganic Dust" – $99,960

AWARDS AND ACCOMPLISHMENTS

[1] 2026 Penn State Astronomy & Astrophysics Research Mentoring Award
[2] 2018 AAS Chambliss Astronomy Student Achievement Award
[3] 2017 Barry M. Goldwater Scholar Honorable Mention
[4] 2017 NSF REU: Northern Arizona University
[5] 2017 USF Undergraduate Research Conference's Excellence in Research Award

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STUDENTS MENTORED

Graduate Students

[1] Lishan Shi (PSU Institute for Computational and Data Sciences – co-advised with Prof. Joel Leja)

Undergraduate Students

[1] William Keck (PSU '28)
[2] Aidan Rowland (PSU '27)
[3] Aidan Groenendaal (PSU '26 → REU at Northwestern → Masters student at DePauw University)
[4] Lennon Nichol (PSU '26 → PSU postbac scholar)
[5] Ana Christina Scigliani (PSU '25 → PSU postbac scholar)
[6] Angelina Estadt (PSU '25 → PhD student at the University of Delaware)
[7] Brock Ewing (BU '25)
[8] Angel Morales (BU '25)

TEACHING EXPERIENCE

Instructor of Record

[1] Penn State University – AST 237W Astronomy Communications – Fall 2025
[2] Boston University – CC 212 Core Natural Science II – Spring 2023
[3] Boston University – CC 111 Core Natural Science I – Fall 2022

Guest Lecturer

[1] Penn State University – Ast 585 Graduate Course in SETI – 4 lectures across Fall 2026

Teaching Assistant

[1] Boston University – AS 101 The Solar System – 7 semesters
[2] Boston University – AS 100 Cosmic Controversies – 1 semester
[3] Boston University – AS 102 The Astronomical Universe – 2 semesters
[4] Boston University – AS 105 Alien Worlds – 1 semester
[5] Boston University – AS 107 Life Beyond Earth – 1 semester
[6] Boston University – AS 202 Principles of Astronomy I – 1 semester
[7] Boston University – AS 203 Principles of Astronomy II – 1 semester
[8] University of South Florida – AST 4721L Astrophysics Laboratory – 1 semester

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HIGH PERFORMANCE COMPUTING EXPERIENCE AND COMPUTING SKILLS

I have extensive experience (>200,000+ Service-Unit hours) in high-performance computing on various clusters: The Massachusetts Green High Performance Computing Center, The Roar Supercomputer (PSU), Monsoon (NAU), TACC Stampede, and CIRCE (USF).

Languages: Python, Fortran90, R, and SQL
Software Packages: Prospector, piXedfit, FSPS, sedpy, GADGET-2, Arepo, TensorFlow.
Paradigms: Stellar population synthesis, simulation-based inference, magnetohydrodynamics modelling, N-body simulations, generative adversarial networks, autoencoders.

OBSERVING EXPERIENCE

* indicates that data were taken by me on-site or remotely.

[1] The 11m Hobby-Eberly Telescope. Instruments: LRS2 & HPF - 35 hours
[2] The 1.8m Perkins Telescope.* Instruments: PRISM - 6 hours
[3] The 4.4m Lowell Discovery Telescope.* Instruments: LMI, DeVeny Spectrograph - 5 hours
[4] The Allan Telescope Array (radio).* Beam-forming mode - 5 hours

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INVITED SCIENTIFIC TALKS

[1] **Curtis, O.,** et al. "A Light in the Dark: Unveiling Delayed Star Formation Histories Within Cosmic Voids." Institute for Gravitation and the Cosmos Neighborhood Meeting, Penn State University, April 2nd, 2026. Oral Presentation.

[2] **Curtis, O.,** et al. "A Light in the Dark: Unveiling Delayed Star Formation Histories Within Cosmic Voids." Tea Talk, California Institute of Technology, March 23rd, 2026. Oral presentation.

[3] **Curtis, O.**, et al. "Void Galaxies and Active Galactic Nucleus Activity in ZOBOV-identified TNG300 Voids from z= 3 to z= 0." Tea Talk, Carnegie Observatory, March 23rd, 2026. Oral presentation.

[4] **Curtis, O.,** et al. "When Galaxies Glow with Industry: A Search for Dyson Sphere Waste Heat from Galaxy-Spanning Civilizations." PANOSETI All Hands Meeting, University of San Diego, March 18th, 2026. Oral Presentation.

[5] **Curtis, O.,** et al. "When Galaxies Glow with Industry: A Search for Dyson Sphere Waste Heat from Galaxy-Spanning Civilizations." Astrobiology Colloquium Series, University of California Riverside, March 13th, 2026. Oral Presentation.

[6] **Curtis, O.**, et al. "Dyson Spheres, Black Holes, and Minds: A Look at the Dyson Minds 2025 Workshop." IAUS 404: Advancing the Search for Technosignatures. March 4th, 2026. Oral presentation. https://www.youtube.com/watch?v=pz6NCJ9dVDI

[7] **Curtis, O.**, et al. "Probing Galaxy Evolution in Cosmic Voids with the TNG300 Simulation and the HETDEX Survey." Café-club Paper Talk. Laboratoire d'Astrophysique de Marseille, January 30th, 2026. Oral Presentation.

[8] **Curtis, O.,** et al. "Overview of Extragalactic Waste Heat SETI." Fall 2025 Penn State Extraterrestrial Intelligence Seminar Series, Penn State University, State College, Pennsylvania, September 25th, 2025. Oral Presentation.

[9] **Curtis, O.,** et al. "Disentangling Technological Waste Heat from Extragalactic Dust." The 2025 PSETI Symposium, The Penn Stater Hotel and Conference Center, State College, Pennsylvania, August 18th – 21st, 2025. Oral Presentation. https://psu.mediaspace.kaltura.com/media/t/1_yb9yss5f

[10] **Curtis, O.,** and Ciardullo, R. "Void Galaxies in the Spitzer/HETDEX Exploratory Large-Area (SHELA) survey." 246th Meeting of the American Astronomical Society, Anchorage, Alaska, June 8th – 12th, 2025. Oral Presentation.

[11] **Curtis, O.**, et al. "Voids and Void Galaxies in the TNG300 Simulation." The Co-evolution of the Cosmic Web and Galaxies Across Cosmic Time, The Kavli Institute for Theoretical Physics, Santa Barbara, California, February 6th – 9th, 2023. Oral Presentation. https://www.youtube.com/watch?v=AnXapqui_R4

[12] **Curtis, O.**, et al. "Voids and Void Galaxies in the TNG300 Simulation." Physics and Astronomy Seminar Series, Tufts University, October 22nd, 2021. Oral Presentation.

PROFESSIONAL SERVICE

[1] **2026 Frontiers in Astronomy and Space Sciences Research Topic on Void Galaxies** – Invited topic coordinator for a collaborative article collection in the journal's Cosmology section

[2] **Penn State Department of Astronomy and Astrophysics Undergraduate Curriculum Committee** – Member for the 2026–2027 academic year

[3] **2026 PULSAR (Postdocs and Undergrads Linked for Summer Astrophysics Research) Program, Penn State Department of Astronomy and Astrophysics** – Postdoc mentor in the department's pilot initiative pairing undergraduates with postdocs on funded 10-week summer research projects

[4] **2025 Penn State Extraterrestrial Intelligence Center Symposium, State College, PA, August 18–21, 2025** – LOC Chair, SOC Member

[5] **Penn State Extraterrestrial Intelligence Center Seminar Series** – Organizer Fall 2024 - present

[6] **Journal Referee** – The Astrophysical Journal (ApJ) and The Astrophysical Journal Letters (ApJL)

[7] **2025 Dyson Minds Workshop, Cambridge, MA, June 3–4, 2025** – Organizing committee member for an interdisciplinary workshop funded by Penn State, MIT, and Harvard

[8] **Physics Pedagogy Journal Club** – Organized and ran a weekly journal club in Spring 2022 at BU focused on physics education research and how to incorporate these ideas into our classrooms.

[9] **Graduate Student Representative** – Liaison between BU graduate students and faculty for the 2021–2022 academic year

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PUBLIC OUTREACH

[1] **Astronomical Society of Harrisburg (ASH) Star Party –** Between the nights of June 21st - 22nd I volunteered as a speaker and telescope operator at ASH's yearly star party.

[2] **Penn State AstroFest –** I volunteered at AstroFest 2025 and 2026, running educational demonstrations and planetarium shows to over a thousand guests over 4 nights.

[3] **Public Observing Nights –** For 15 clear nights a year over 6 years, I set up telescopes and managed the observing session at Boston University's Judson B. Coit Observatory's public open nights.

[4] **Skype a Scientist** – Since Spring 2019, I have participated in the Skype a Scientist program. I have talked with over 25 K-12 classrooms across the country, giving mini-lessons on space-based topics and talked to them about life as an astronomer.

[5] **USF Engineering EXPO** – I organized USF's Society of Physics Students to perform multiple 45-minute shows at USF's annual Engineering EXPO, a 3 day event that brings together thousands of Tampa Bay public school students. Years performed: 2016, 2017, and 2018.

PROFESSIONAL DEVELOPMENT

[1] **Teaching evaluation, PSU Grove Center for Excellence in Teaching Education** – Incorporated feedback from Grove Center and astronomy teaching faculty during Fall 2025 for my course AST 237W Astronomy Communications.

[2] **BU's Center for Teaching & Learning's Inclusive Pedagogy Workshop** – Attended the seminar series of dialogue-based inclusive teaching practices from June 11–13, 2024.

[3] **Teaching evaluation, BU Core Curriculum** – Incorporated feedback from BU Core Curriculum Master Lecturer Robin Stevens during three lectures of CC 212 Core Natural Sciences II in Spring 2023.

First- and Second-Author Publications

Link to ADS Library: https://ui.adsabs.harvard.edu/user/libraries/ylWhhdzqTJOlPUolGK-r1w

[1] **Curtis, O.**, et al. "A Spectral Framework for Testing the Quasi-Star Hypothesis in Little Red Dots I: Weighing LRDs by Their Super-Eddington Luminosity Ratios—No Signs of Overmassive Black Holes." *The Astrophysical Journal.* (submitted.) 10.48550/arXiv.2609.09265

[2] **Curtis, O.**, et al. "The $\hat{G}$ Infrared Search for Extraterrestrial Civilizations with Large Energy Supplies. V. When Galaxies Glow with Industry." *The Astrophysical Journal.* (submitted.) 10.48550/arXiv.2608.12458

[3] **Curtis, O.**, McDonough, B., and Brainerd, T. G. "A High-Mass Size Deficit for Void Galaxies in the Complete SDSS DR7." *The Astrophysical Journal.* (accepted.) 10.48550/arXiv.2609.19253

[4] **Curtis, O.**, et al. "The Dyson Minds 2025 Workshop: SETI Around Black Holes." *Publications of the Astronomical Society of the Pacific.* 138.4 (2026): 046001. doi: 10.1088/1538-3873/ae5a02

[5] **Curtis, O.**, et al. "Void Galaxies and Active Galactic Nucleus Activity in ZOBOV-identified TNG300 Voids from z= 3 to z= 0." *The Astrophysical Journal.* 998.2 (2026): 325. doi: 10.3847/1538-4357/ae2f48

[6] McDonough, B., **Curtis, O.**, and Brainerd, T.G. "Measuring Scaling Relations: Fitting Technique Matters." *The Astrophysical Journal Letters.* 986.2 (2025): L32. doi: 10.3847/2041-8213/ade13c

[7] **Curtis, O.**, McDonough, B., and Brainerd, T. G. "Density Profiles of TNG 300 Voids across Cosmic Time." *The Astrophysical Journal.* 985.2 (2025): 244. doi: 10.3847/1538-4357/adcebf

[8] McDonough, B., **Curtis, O.,** et al. "Intrinsic and Environmental Effects on the Distribution of Star Formation in TNG100 Galaxies." *The Astrophysical Journal.* 978.1 (2024): 67. doi: 10.3847/1538-4357/ad9747

[9] **Curtis, O.**, McDonough, B., and Brainerd, T. G. "Properties of Voids and Void Galaxies in the TNG300 Simulation." *The Astrophysical Journal* 962.1 (2024): 58. doi: 10.3847/1538-4357/ad18b4

[10] McDonough, B., **Curtis, O.**, and Brainerd, T. G. "Resolved Star Formation in TNG100 Central and Satellite Galaxies." *The Astrophysical Journal* 958.1 (2023): 19. doi: 10.3847/1538-4357/acfe0b

[11] **Curtis, O.**, Brainerd, T. G., and Hernandez, A. "Cosmic voids in GAN-generated maps of large-scale structure." *Astronomy and Computing* 38 (2022): 100525. doi: 10.1016/j.ascom.2021.100525

[12] Womack, M., **Curtis, O.**, et al. "The Visual Lightcurve of Comet C/1995 O1 (Hale–Bopp) from 1995 to 1999." *The Planetary Science Journal* 2.1 (2021): 17. doi: 10.3847/PSJ/abd32c
   a. Software Release: **Curtis O.**, et al. "ICQSplitter: Python Tool for Handling Visual or CCD Magnitudes of Comets v3.0," *Zenodo*, (2020) doi: 10.5281/zenodo.3628044

[13] **Curtis, O.**, and Brainerd, T. G. "Fast generation of large-scale structure density maps via generative adversarial networks." *Research Notes of the AAS* 4.6 (2020): 90. doi: 10.3847/2515-5172/ab9e01

[14] Chandler, C. O., **Curtis, O.**, et al. "SAFARI: Searching asteroids for activity revealing indicators." *Publications of the Astronomical Society of the Pacific* 130.993 (2018): 114502. doi: 10.1088/1538-3873/aad03d

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Books

[1] **Curtis, O.**, and Wright, J. T. "Dyson Minds: Large-Scale Postbiological Intelligences in Dyson Spheres." In *The Postbiological Universe: Artificial Intelligence in a Cultural Cosmos*, edited by S. J. Dick and M. Lupisella. Cambridge University Press. (invited chapter, in preparation.)

Co-Authored Publications

[1] Chávez Ortiz, Ó. A., et al. (incl., **Curtis, O.**) "Euclid + HETDEX: Tracing observables with neural estimators (TONE)." *Astronomy & Astrophysics.* (submitted.)

[2] Garner, R., et al. (incl., **Curtis, O.**) "Mapping the Stellar Fossil Record of M101 with HETDEX: Spatially Resolved Ages and Metallicities." *The Astrophysical Journal.* (submitted.)

[3] Zackrisson, E., et al. (incl., **Curtis, O.**) "Project Hephaistos – IV. James Webb Space Telescope Observations of Two Dyson Sphere Candidates." *Monthly Notices of the Royal Astronomical Society.* (submitted.) 10.48550/arXiv.2607.09460

[4] Mirza Khanlari, M., et al. (incl., **Curtis, O.**) "Population III Host Candidates at z~2: Strong He II λ1640 and Absent UV Metal Lines in HETDEX Lyα Emitters." *The Astrophysical Journal.* (accepted.) 10.48550/arXiv.2609.16134

[5] Moon, J. et al. (incl., **Curtis, O.**) "HETDEX [OII] galaxies at z≤0.48: Volume-limited samples and their power spectra." *The Astrophysical Journal.* (accepted.) 10.48550/arXiv.2607.08453

[6] Cleri, N., et al. (incl., **Curtis, O.**) "RUBIES: The Evolution of the Ionization Parameter from $0 < z < 9$." *The Astrophysical Journal.* (accepted.) 10.48550/arXiv.2605.30410

[7] Mentuch Cooper, E., et al. (incl., **Curtis, O.**) "HETDEX Public Data Release 1: Source Catalog 2 and Datacubes from ~90 sq. degrees of Integral-Field Optical Spectroscopy." *The Astrophysical Journal Supplement Series.* 284.2 (2026): 67. doi: 10.3847/1538-4365/ae6068

[8] Burnham, E., et al. (incl., **Curtis, O.**) "It's More Complicated Than You Think: A Forward Model to Infer the Recent Star Formation History, Bursty or Not, of Galaxy Populations." *The Astrophysical Journal.* 1001.2 (2026): 205. doi: 10.3847/1538-4357/ae562d

[9] Lujan Niemeyer, M., et al. (incl., **Curtis, O.**) "Lyα Intensity Mapping in HETDEX: Galaxy-Lyα Intensity Cross-power Spectrum." *The Astrophysical Journal.* 999.2 (2026): 177. doi: 10.3847/1538-4357/ae3a98

MANUSCRIPTS IN PREPARATION

In order of completion estimate
* Indicates a direct mentee

[1] **Curtis, O.**, et al. "A Spectral Framework for Testing the Quasi-Star Hypothesis in Little Red Dots II: A Chromosphere and a Metastable Helium Wind Above the Photosphere." (in preparation.)

[2] Shi, L.,* **Curtis, O.,** et al. "Millions of Galaxies But No Time: Rapid Inference of Galaxy Properties with Neural Density Estimators." (in preparation.)

PUBLIC TALKS

[1] **Curtis, O.** "A Brief History and Current Search for Extraterrestrial Intelligence Research." Cherry Springs State Park, Potter County, Pennsylvania, June 21st, 2025. Oral Presentation, ~150 guests.

[2] **Curtis, O.** "A tour of the Galaxy." Judson B. Coit Public Observatory Night, Boston, Massachusetts, April 19th, 2023. Oral Presentation, ~40 guests.

Press Releases

[1] Collins, M., and Buckley, N. "Episode 44 – asteroid impacts, chemical evolution and quasi-stars." *StarXiv Podcast* (2026).

[2] Blue Marble Space. "Minds Around the Abyss: What Would an AI Living on a Black Hole Look Like? Olivia Curtis on Dyson Minds and Black Hole Technosignatures for IAUS404." *Planetary Perspectives* (2026).

[3] Anifowose, J. "Scientists say aliens may be harnessing energy from black holes." *Geekspin* (2026).

[4] Cost, B. "Scientists think alien civilizations could be using 'Dyson Spheres' to siphon energy from black holes." *The New York Post* (2026).

[5] Starr, M. "JWST Just Examined Two Stars Suspected of Harboring Dyson Spheres. Here's What It Found…" *ScienceAlert* (2026).

[6] MacDonald, F. "Experts Met at MIT to Discuss The Possibility of Alien Tech Around Black Holes. Here's What Was Discussed." *ScienceAlert* (2026).

[7] Michael, J. "Here's a look at the Penn State festival that kicks off after Arts Fest booths close." *The Centre Daily Times* (2025).

CONTRIBUTED TALKS AND POSTERS

* Indicates a direct mentee

[1] Nichol, L.,* Wright, J. T., **Curtis, O.**, and the NEID Science Team. "A Real Piece of Sky: Chromatic Aberration and the NEID Spectrograph." 247th Meeting of the American Astronomical Society, Phoenix, Arizona, January 4th – 8th, 2026. Poster.

[2] Groenendaal, A.,* **Curtis, O.**, Wright, J. T., and Tusay, N. "Proof of Concept: Lunar Occultations as an RFI Rejector." 247th Meeting of the American Astronomical Society, Phoenix, Arizona, January 4th – 8th, 2026. Poster.

[3] Rowland, A.,* **Curtis, O.**, and Wright, J. T. "Advancing the Ĝ Survey: Multiwavelength Analysis of Galaxies in the Search for Advanced Civilizations Beyond the Milky Way." 247th Meeting of the American Astronomical Society, Phoenix, Arizona, January 4th – 8th, 2026. Poster.

[4] Scigliani, A.,* Wright, J. T., **Curtis, O.**, and Tusay, N. "New Pixel-level K2 Detrending Method and The Random Transiter (EPIC 249706694/HD 139139)." 247th Meeting of the American Astronomical Society, Phoenix, Arizona, January 4th – 8th, 2026. Poster.

[5] Groenendaal, A.,* **Curtis, O.,** et al. "Proof of Concept: Lunar Occultations as an RFI Rejector." 2025 Penn State Undergraduate Research Exhibition, 321 Boucke Building, State College, Pennsylvania, April 14th – 16th, 2025. Poster.

[6] Nichol, L.,* **Curtis, O.,** et al. "A Real piece of Sky: Chromatic Atmospheric Effects and the NEID Spectrograph." 2025 Penn State Undergraduate Research Exhibition, 321 Boucke Building, State College, Pennsylvania, April 14th – 16th, 2025. Poster.

[7] Scigliani, A.,* Wright, J. T., Tusay, N., and **Curtis, O.** "Demystifying The Random Transiter." 245th Meeting of the American Astronomical Society, January 12th – 16th, 2025. Poster.

[8] McDonough, B., **Curtis, O.**, et al. "Exploring IllustrisTNG simulations to derive observationally comparable star formation rates and metallicities." Recipes to Regulate Star Formation at All Scales:

From the Nearby Universe to the First Galaxies, Baltimore, Maryland, April 15th – 19th, 2024. Workshop.

[9] **Curtis, O.** and Brainerd, T. G. "Fast Generation of Large-Scale Structure Density Maps via Generative Adversarial Networks." 236th Meeting of the American Astronomical Society, Virtual Conference, June 1st – 3rd, 2020. Interactive Poster Presentation.

[10] N. Lastra, **Curtis, O.**, et al. **"**Best Secular Lightcurve for Comet Hale-Bopp and Implications for Dust Production Rates." American Physical Society March Meeting, Los Angeles, California, March 5th – 9th, 2018. Poster Presentation.

[11] **Curtis, O.**, et al. "SAFARI: Searching Asteroids for Activity Revealing Indicators." 232nd Meeting of the American Astronomical Society, Denver, Colorado, June 3rd – 7th, 2018. Interactive Poster Presentation.

[12] Womack, M., Lastra, N., **Curtis, O.**, et al. "The Secular Lightcurve and Photometric Evolution of C/1995 O1 (Hale-Bopp)." 49th American Astronomical Society Annual Division of Planetary Science Meeting, Provo, Utah, October 15th – 20th, 2017. Interactive Poster Presentation.

[13] Harrington, O., Womack, M., **Curtis, O.**, et al. "Correlation Between Cometary Gas/Dust Ratios and Heliocentric Distance." 49th American Astronomical Society Annual Division of Planetary Science Meeting, Provo, Utah, October 15th – 20th, 2017. Poster Presentation.